PRESENTATION SCRIPT — TWO SPEAKERS
"Development of an Image-based Disease Detection Leveraging AI/ML Techniques and Computer Vision"
Source deck: Completion PPT - Presentation.pptx.pdf (19 slides)
Speakers: AAKASH (Aakash Trimbak Jawle) and KAMAAKSHA (Kamaaksha Rajendra Kharul)
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Split logic: Aakash opens with project framing (intro, objectives, benefits) and owns the "model/data engineering" slides (CNN baselines, soybean, knowledge-grounded pipeline, dataset/training config, results-at-a-glance). Kamaaksha takes over from the timeline/overview slide and owns the "evaluation and outcome" slides (VLM selection, PlantDx pipeline, internal test, cross-domain evaluation, publication). Both close together. Total runtime target: ~14-16 minutes at a normal conversational pace.
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SLIDE 1 — TITLE
Speaker: AAKASH

Good [morning/afternoon] everyone. I'm Aakash Trimbak Jawle, and with me is my co-fellow Kamaaksha Kharul. We're presenting our CHANAKYA Fellowship project at TIH IIT Bombay, titled "Development of an Image-based Disease Detection System Leveraging AI/ML Techniques and Computer Vision," carried out under the mentorship of Dr. Santwana Gudadhe at Pimpri Chinchwad College of Engineering, Pune. Over the next few minutes, we'll walk you through what we set out to do, how the work progressed, what we built, and what we found — including where it worked well and where it didn't.

SLIDE 2 — PROJECT CARD
Speaker: AAKASH

Quickly, the project ran from 1st December 2025 to 30th September 2026. Financially, the total sanctioned amount was 2.40 lakh rupees, of which 1.92 lakh has been disbursed from TIH — that's 1.20 lakh and 0.96 lakh per fellow respectively, with a balance of 0.48 lakh still pending. On the recognition side, our research paper has been accepted for publication in the Proceedings of ICAIN 2026, Springer's LNNS series, and the acceptance notification placed it among the top 20% of papers reviewed. We didn't pursue any IP or patent filings, and there's been no commercialization — the outputs here are research workflows and trained adapters, and field validation on independently collected images remains a prerequisite before any real deployment is considered.

SLIDE 3 — BACKGROUND & PROBLEM STATEMENT
Speaker: AAKASH

So why this problem? Crop diseases and pests often look visually similar to each other, which makes manual diagnosis slow and inconsistent — and that's true across all three crops we worked with: tomato, mango, and soybean. Each of these has its own dataset and its own pipeline in our project — TIH and PlantVillage data for tomato, PlantDx for mango, and SoyNet plus TIH data for soybean. Expert visual inspection doesn't scale well, and it's entirely dependent on expert availability. What really motivated the technical direction, though, was this: when we ran a general-purpose vision-language model zero-shot — no fine-tuning at all — it only managed 7.3% accuracy on tomato and 17.5% on mango. So our objective became clear: adapt CNNs and vision-language models parameter-efficiently, using QLoRA, on workstation-class hardware, and be honest about how well these models generalize beyond their own training data.

SLIDE 4 — OBJECTIVES
Speaker: AAKASH

That translated into eight concrete objectives. First, investigate multiple VLM families and select a backbone for crop diagnosis. Second, curate and audit our datasets — verifying images, removing duplicates and label conflicts, and freezing our splits so nothing leaks between train, validation, and test. Third, apply parameter-efficient fine-tuning — specifically adapting Qwen2.5-VL-7B-Instruct using QLoRA on a single 24 GB GPU. Fourth and fifth, train and evaluate dedicated classifiers — an 11-class tomato model and an 8-class mango model. Sixth, measure generalization honestly on an independent external benchmark, not just our own test set. Seventh, build a knowledge-grounded diagnosis layer — a cited disease knowledge base, an ontology, and a controlled vocabulary. And eighth, make sure everything is reproducible: release tested, config-driven code, and publish the findings.

SLIDE 5 — EXPECTED BENEFITS OF THE PROJECT
Speaker: AAKASH

In terms of benefits: this gives us automated disease and pest identification that reduces how much we depend on manual inspection. It demonstrates efficient use of large VLMs — we're only training 40.37 million LoRA parameters, with a peak memory footprint of 10.57 GB on a single 24 GB GPU, not the full 7-billion-parameter model. Our data audit removed 550 duplicate or contradictory files, so no image appears in two splits — that's cleaner, leak-free data. The pipelines themselves are config-driven, with fixed seeds, split hashes, preflight checks, and tests — so they're reusable and reproducible. Running internal versus external evaluation gives us an honest robustness assessment that pinpoints exactly which classes are weak, like Late Blight, which we'll come back to. And the same recipe we used for tomato and mango is extensible to more crops going forward. Put simply: a reproducible, parameter-efficient VLM workflow with transparent evaluation that tells us what more data we'd need before any

field deployment.

Now I'll hand it over to Kamaaksha, who'll take you through how this work actually progressed month by month, and then into the evaluation results.

SLIDE 6 — OVERALL WORK PROGRESS & MILESTONE TIMELINE
Speaker: KAMAAKSHA

Thanks, Aakash. Let me walk you through the timeline — eight milestones across the ten months of the fellowship. In December and January, we laid the foundation: image preprocessing, quality checks, augmentation, and our first custom CNN trained from scratch, alongside EfficientNetB3, MobileNetV2 and ConvNeXt-Tiny — the custom CNN reached 98.51% on tomato. In January-February, we extended this to soybean with ConvNeXt-Tiny on both Keras and PyTorch, SoyNet CNN experiments, and TIH VDI training — ConvNeXt-Tiny PyTorch hit 99.92%, and SoyNet ResNet50 hit 95.16%. From March to June, we screened vision-language models — LLaVA and BLIP-2 LoRA experiments, Qwen2.5-VL-7B fine-tuning, and a seven-model zero-shot benchmark — Qwen2.5-VL came out on top at 98.50% accuracy. Between May and July, we built PlantDx and our knowledge-grounding layer — a disease knowledge base, an ontology, and a 1,966-caption corpus. In July, we ran the full knowledge-grounded QLoRA pipeline on tomato and mango — tomato jumped from 7.3% to 93.7%, and mango from 17.5% to 82.0%. Through August-September, we moved to our final TIH dataset — cleaning and deduplicating 7,095 images down to 6,545, and training Qwen2.5-VL-7B with QLoRA. In September, we evaluated — internal held-out accuracy of 92.76%, and an external 176-image test at 69.32%. And by October, the paper was accepted — published in Springer's LNNS series. That's the whole arc, from CNN baselines through VLM exploration, PlantDx, knowledge-grounded QLoRA, cross-domain evaluation, and finally publication — with project closure on 30th September 2026.

SLIDE 7 — PROJECT OVERVIEW & TECHNICAL PROGRESSION
Speaker: KAMAAKSHA

To zoom out for a second: the technical journey ran through ten stages — foundation, CNN baselines, VLM selection, knowledge-grounded diagnosis, TIH dataset adaptation, QLoRA fine-tuning, validation, internal testing, cross-domain evaluation, and publication — grouped into three phases: foundation and research, model development, and evaluation and publication. The core problem we tackled is image-based crop disease and pest diagnosis in the AgriTech domain, where manual inspection is slow and expert-dependent. Our approach combined vision-language models and CNNs, adapted parameter-efficiently, on workstation-class GPU hardware and, for part of the work, Apple Silicon using MLX. Crucially, we insisted on evaluating ourselves honestly — beyond our own training data, through an independent cross-domain test. Across all three crops: for tomato, we covered both CNN baselines and VLM tracks, on PlantVillage and our own

TIH 11-class dataset, landing at 92.76% held-out and 69.32% externally, and that's the work now published at ICAIN 2026. For mango, we built an 8-class knowledge-grounded VLM on PlantDx, taking it from 17.5% zero-shot to 82.0% after QLoRA fine-tuning. For soybean, SoyNet CNN baselines across Keras and PyTorch reached 95.16% test accuracy with ResNet50. And in terms of what we're leaving behind: seven GitHub repositories under Apache-2.0, trained CNN and QLoRA adapter checkpoints, audited and deduplicated datasets with frozen splits, full evaluation reports and confusion matrices, a Streamlit demo application, and the accepted ICAIN 2026 paper itself.

SLIDE 8 — TOMATO: CNN BASELINES (PLANTVILLAGE)
Speaker: AAKASH

Let's get into the technical detail, starting with our tomato CNN baselines on PlantVillage. We trained five architectures in total. The headline numbers: ConvNeXt-Tiny in PyTorch reached 99.92% on a genuinely leak-free 70/15/15 split with 2,411 test images. Our custom CNN, trained from scratch, reached 98.51% — but I want to flag that this one is indicative only, because its 80/20 split was drawn on the same machine and largely overlaps the training data. ConvNeXt-Tiny in Keras, evaluated on a separate held-out validation set of 3,198 images, reached 96.15%. All models used a 224-by-224 input size and trained on an RTX 4500 Ada GPU. If you look at the chart, the progression runs from MobileNetV2 at 90.9% up through EfficientNetB3, ConvNeXt in Keras, the custom CNN, and finally ConvNeXt in PyTorch at 99.9%. The key takeaway is that ConvNeXt-Tiny PyTorch, with its disjoint split, is the more credible benchmark here — at 99.92% accuracy, Macro-F1 0.9991 — not the custom CNN's 98.51%. We also built a crop-validation pipeline — using HSV green-pixel ratio, edge ratio, and cosine similarity to MobileNetV2 embeddings — to gate inference before prediction, so the system can reject obviously non-crop images.

SLIDE 9 — SOYBEAN: CNN CLASSIFICATION (SOYNET & TIH VDI)
Speaker: AAKASH

Moving to soybean. We started with 3,720 camera-click images from the SoyNet dataset — 3,164 diseased and 556 healthy — resized to 256-by-256, with augmentation applied only to the training data. We used a stratified 70/15/15 split with a fixed random seed, giving us 2,603 training, 557 validation, and 560 test images. We trained this in two independent runs — first in Keras, then in PyTorch on 27th January 2026 — and trained six backbones in total: ResNet50 and MobileNetV2 on both frameworks, plus EfficientNetB0 and EfficientNetB3 on PyTorch. I should flag one limitation here: a separate TIH-VDI soybean model does exist conceptually, but it's not included in these results, because the VDI environment was unavailable to us. There's a strong class imbalance in this dataset — roughly 5.7 diseased images for every healthy one — so the precision, recall and F1 figures we report are specifically for the healthy, minority

class. The best overall performer was ResNet50 in Keras, at 95.16% accuracy with a ROC-AUC of 0.9903. MobileNetV2 in Keras came in at 89.07%. On the PyTorch side, ResNet50, MobileNetV2, EfficientNetB0 and EfficientNetB3 all achieved perfect recall of 1.0, with EfficientNetB3 the strongest PyTorch run at 93.93% accuracy.

Over to Kamaaksha for the vision-language model work.

SLIDE 10 — TOMATO: VLM MODEL SELECTION
Speaker: KAMAAKSHA

Thanks. Before we committed to a VLM backbone, we screened several candidates. We started with LLaVA 1.5 and 1.6, but hit a batch-collation incompatibility and had to discontinue that track in March. We then tried BLIP-2 with OPT-2.7B, which learned a structured output format and reached 89.00% accuracy. We attempted Florence-2 next, but ran into environment and compatibility failures in April, and again when we retried it in June — so that was discontinued too. Qwen2.5-VL-7B-Instruct was selected as our backbone in April, reaching 98.50% accuracy right away. We then used it for a restructured 15-class PlantDx run in May, at 98.00% accuracy, and ran a seven-model zero-shot benchmark on Apple Silicon using MLX-VLM in June. If you look at the screening chart: LLaVA managed only 2%, Florence-2 effectively 0%, BLIP-2 89%, and Qwen2.5-VL 98.5%. The important point is that every one of these models was screened under exactly the same PlantVillage structured-caption protocol, so the comparison is fair — and LLaVA and Florence-2 simply didn't produce usable output, so they were dropped. Qwen2.5-VL-7B-Instruct's reliability — zero unparseable outputs across every run — is what made it the backbone for every later track: PlantDx and, eventually, our final TIH tomato model.

SLIDE 11 — PLANTDX & TOMATO PIPELINE
Speaker: KAMAAKSHA

Before we built the full knowledge-grounded pipeline, we had an intermediate milestone: the PlantDx 15-class restructure in May 2026, fine-tuned with Qwen2.5-VL LoRA, run ID run_001, on Apple Silicon via MLX. On a frozen held-out split of 200 PlantVillage images, this reached 98.00% accuracy and a Macro-F1 of 0.9777. We also built a Streamlit inference application — version 1.0.0 — that compares base-model versus fine-tuned predictions on that frozen test split, and we used it for live demonstration during this review. To be clear, this is a research prototype, not a production deployment. This 15-class result is best understood as a stepping stone — it's what we built before the full knowledge-grounded corpus and QLoRA pipeline that Aakash will take you through next.

Back to Aakash for the knowledge-grounded pipeline.

SLIDE 12 — KNOWLEDGE-GROUNDED TOMATO & MANGO PIPELINE
Speaker: AAKASH

This is where the biggest accuracy gains came from. We built a disease knowledge base covering 18 cited conditions — 10 for tomato, 8 for mango — compiled into a typed, evidence-linked ontology, backed by a controlled vocabulary and symptom lexicon. From there, we generated a caption corpus deterministically, through concepts and templates, validated across multiple stages — critically, there's no VLM or LLM involved anywhere in that caption-generation path, and no image analysis either; everything traces back to cited knowledge. We applied this to audited, normalized images from PlantVillage for tomato and MangoLeafBD for mango, and then ran QLoRA fine-tuning — Qwen2.5-VL-7B in 4-bit, rank 16, on Apple Silicon. The results speak for themselves: tomato went from 7.3% zero-shot to 93.7% after fine-tuning, on 910 test images, with a Macro-F1 of 0.919 and Weighted-F1 of 0.937. Mango went from 17.5% to 82.0%, on 200 test images, with both Macro and Weighted F1 at 0.811. We held ourselves to a few principles throughout — cited knowledge only, observability- and severity-honest language, and bit-for-bit reproducibility. One important caveat: these are in-distribution scores on public, single-leaf datasets — they're separate from, and not directly comparable to, our TIH benchmark, which Kamaaksha will cover shortly.

SLIDE 13 — FINAL TIH TOMATO DATASET & TRAINING
Speaker: AAKASH

For our final, most rigorous track, we started with 7,095 raw labelled images across 11 class folders. We ran every image through verification — checking for supported extensions and that the file actually decodes — then applied SHA-256 exact deduplication, collapsing same-class copies down to one, and removed contradictory duplicates, where the same image hash appeared under more than one label. That left us with 6,545 retained images — 92.25% of the original set — split in a class-stratified 80/10/10 ratio with a fixed seed, giving 5,247 training, 649 validation, and 649 test images, with the split hashes frozen so they can never silently change. There's a strong class imbalance here too — Late Blight has 195 test images versus just 9 for Healthy Fruit — so we used inverse-square-root sampling to compensate. On the training side: our base model is Qwen2.5-VL-7B-Instruct with a frozen vision encoder, quantized to 4-bit NF4 with double quantization and BF16 compute. We used LoRA adapters at rank 64, alpha 128, dropout 0.05, applied only to the query, key, value and output projections — that's 40.37 million trainable parameters. We optimized with Paged AdamW 8-bit, a learning rate of 4e-5 on a cosine schedule with 3% warm-up, an effective batch size of 8 through gradient accumulation, and planned for 3 epochs — we actually completed 2.909. Checkpoint selection prioritized validation macro-F1, then accuracy, then loss, with early stopping at a patience of 1.

Over to Kamaaksha for how this model actually performed.

SLIDE 14 — EXPERIMENTAL RESULTS: INTERNAL TEST (TIH, 649 IMAGES)
Speaker: KAMAAKSHA

On our internal, held-out TIH test set of 649 images, the final model achieved 92.76% accuracy — that's 602 out of 649 correct — with a Macro-F1 of 93.78%, which we treat as our primary metric given the class imbalance. Macro recall came in at 95.57%, against a macro precision of 92.71%, and inference latency was 0.327 seconds per image on a single RTX 4500 Ada GPU. Training took 1,908 optimizer steps over 2.909 epochs, roughly 192.7 minutes, with a peak memory footprint of 10.57 GB — and notably, zero unparseable model outputs across the entire test set. Looking at the per-class breakdown: Late Blight, with 195 test images, scored an F1 of 90.91% and recall of 84.62% — its main confusions were with Target Leaf Spot, Leaf Curl and Leaf Miner. Target Leaf Spot had perfect recall but only 68.75% precision, giving it the lowest F1 on this test at 81.48%. And Healthy Fruit scored 100%, but on only 9 test images, so that number comes with very limited statistical support.

SLIDE 15 — CROSS-DOMAIN EVALUATION (INTERNAL VS EXTERNAL)
Speaker: KAMAAKSHA

This is the evaluation we think matters most, because it's the honest test of generalization. We took the same model and ran it against an independent external benchmark — 176 images across 10 of our 11 classes. Internal accuracy was 92.76%; external accuracy dropped to 69.32% — a gap of 23.44 percentage points. Macro-F1 fell from 93.78% to 71.95%, a drop of 21.83 points, and latency rose slightly too, from 0.327 to 0.373 seconds. Looking at per-class F1 across the ten shared classes, the picture is class-dependent rather than uniform: Late Blight collapsed from 90.91% to just 14.29% — 11 of 20 external Late Blight images were misclassified as Target Leaf Spot — while Mealy Bugs and White Fly held up well, at 93.33% and 91.67% respectively. The honest conclusion we draw from this is that cross-domain evaluation revealed a substantial generalization gap, and it underlines the importance of broader, more diverse training data before any real-world deployment.

SLIDE 16 — RESULTS AT A GLANCE: ALL EXPERIMENTS
Speaker: AAKASH

Let me pull all of this together, because we want to be very careful here: this table is not a leaderboard. These nine results come from four different datasets — TIH tomato, PlantVillage, MangoLeafBD and SoyNet — multiple different protocols, from zero-shot to LoRA to QLoRA to standard CNN transfer learning — and two different hardware paths, CUDA on the RTX 4500 Ada and Apple Silicon via MLX. So every row needs to be read in its own experimental context. To summarize the nine: our custom CNN hit 98.51%

on PlantVillage, though that figure is indicative since the split overlaps training; ConvNeXt-Tiny reached a genuine 99.92% on a leak-free split; Qwen2.5-VL zero-shot screening hit 98.50%; the PlantDx 15-class restructure hit 98.00%; our knowledge-grounded QLoRA pipeline took tomato to 93.7% and mango to 82.0%, each starting from a single-digit or high-teens zero-shot baseline; SoyNet ResNet50 reached 95.16% on soybean; and our final TIH model reached 92.76% held-out and 69.32% externally. Three things we'd ask you to keep in mind: PlantVillage is not the same as TIH tomato — one is lab-condition images, the other is our own field and fruit dataset, so please never average those accuracies together. Every zero-shot-to-fine-tuned improvement we show is paired against its own un-tuned baseline, not a competing model's score. And that internal-to-external drop, 92.76% down to 69.32%, is the real headline limitation of this work — not an outlier to explain away.

SLIDE 17 — PUBLICATION, MILESTONE STATUS & LIMITATIONS
Speaker: KAMAAKSHA

On the publication side: our paper, "Efficient QLoRA Fine-Tuning of Qwen2.5-VL for Multi-Class Tomato Disease Classification and Cross-Domain Evaluation," has been accepted at ICAIN 2026, to appear in Springer's LNNS series, Submission ID 953. The authors are Aakash Trimbak Jawle, myself, Dr. Santwana Gudadhe, Harish Lokawad and Rutuja Chavan, and registration was completed on 20th October 2026. The acceptance notification placed it among the top 20% of papers reviewed, with indexing expected in zbMATH, Scopus and DBLP — DOI, volume and page numbers aren't assigned yet. In terms of overall status: we completed CNN and VLM baselines, QLoRA fine-tuning, and both internal and external evaluation, across all three crops, and published the work. On limitations: our PlantVLM caption-retrieval approach was exploratory, tested on only 30 images; Florence-2 was discontinued due to compatibility issues; the separate TIH-VDI soybean model remains unavailable; and, as we discussed, external accuracy sits 23.44 percentage points below internal — that's a real domain gap we're not glossing over. And to close the loop on funding: 2.40 lakh sanctioned, 1.92 lakh disbursed, 0.48 lakh balance still pending, no IP or patent filings, over a fellowship period from 1st December 2025 to 30th September 2026.

SLIDE 18 — CONCLUSION & FUTURE WORK
Speaker: KAMAAKSHA (with AAKASH joining for the close)

KAMAAKSHA: To conclude — we built and evaluated CNN and vision-language model workflows across tomato, mango and soybean, for disease and pest classification, going all the way from audited data to a QLoRA-adapted Qwen2.5-VL-7B model and transparent internal and cross-domain evaluation — and the tomato work has been accepted at ICAIN 2026. Five takeaways: what we built — audited datasets across three crops, CUDA and MLX QLoRA pipelines, a full evaluation framework, and the knowledge-grounded PlantDx

track. Our measurable results — 92.76% internal accuracy and 93.78% macro F1 on TIH tomato, 93.7% and 82.0% on knowledge-grounded tomato and mango, and 95.16% on SoyNet soybean. Our publication at ICAIN 2026. Our key limitation — a 69.32% external accuracy, with Late Blight F1 falling to 14.29%, a clear and acknowledged domain gap.

AAKASH: And our next direction — real-field validation on independently collected images, expanding to more datasets and crops, optimizing for deployment, and improving reliability through further knowledge-grounding.

BOTH: Thank you.

SLIDE 19 — THANK YOU
Speaker: BOTH

Thank you all for your time. We're happy to take any questions.
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END OF SCRIPT